

ROUSHAN KUMAR

Python Systems Developer | FastAPI, Asyncio & Automation Specialist

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PROFESSIONAL SUMMARY

Python Systems Developer specializing in FastAPI microservices, asynchronous execution runtimes, and automated parsing pipelines. Experienced in constructing subprocess-isolated sandboxes, Tree-Sitter code graphs, ChromaDB vector stores, and automated ETL data ingestion scripts. Focused on writing clean, type-hinted, unit-tested Python code to deliver resilient developer tooling and low-latency API services.

TECHNICAL SKILLS

Python Ecosystem: Python 3.x • FastAPI • Asyncio • Subprocess • Tree-Sitter • ChromaDB • PyTest

Data Engineering & ETL: SQLite3 • Pandas • CSV Ingestion • 3NF Schema Design • Bitwise Optimization

Backend & APIs: RESTful Microservices • Node.js • Express.js • C# (.NET Core) • WebSockets • JWT Auth

Languages: C • Rust • Python • TypeScript • JavaScript • SQL • C# • Java

DevOps & Testing: PyTest • Docker • CI/CD • GCP • Git • Vercel • Render • Linux Shell

Security & Architecture: Subprocess Isolation • Import Interception • AES-GCM • PBKDF2 • SOLID Principles

TECHNICAL PROJECTS

Astra Vision - Developer Sandbox & Code Graph Parser (FastAPI, Python, Monaco, Tree-Sitter)

[Live Demo](#) | [GitHub](#)

- Product Architecture:** Architected an automated code-parsing platform and browser sandbox that allows developers to safely execute untrusted code while visualizing full-stack repository dependency graphs.
- Sandbox Runtime Challenge:** Built a self-healing Python execution engine with subprocess isolation and security checkpoints to intercept unauthorized OS file system calls and network socket requests in real time.
- Indexing Performance:** Integrated Tree-Sitter AST compilers and ChromaDB vector search to parse repository syntax nodes, enabling instant code graph dependency analysis under 200ms.

IDBI FinSync - AI-Powered Wealth & Financial Management Engine (Next.js, React, Fastify, Gemini API)

[Live Demo](#) | [GitHub](#)

- Product Architecture:** Co-created an intelligent personal financial management web app in a monorepo that unifies bank ledgers, investment portfolios, and expense streams into a live interactive dashboard.
- AI Integration Challenge:** Embedded "Mitra," an interactive AI wealth consultant leveraging Gemini API to analyze spending habits, detect budget anomalies, and deliver personalized financial guidance.
- Database & Concurrency:** Designed PostgreSQL transaction ledgers validated by Zod schemas and Fastify microservice endpoints, handling concurrent balance updates with zero data loss.

Alyra Lock - Secure Zero-Knowledge Password Vault (React, TypeScript, Express, MongoDB)

[Live Demo](#) | [GitHub](#)

- Product Architecture:** Engineered a client-side zero-knowledge password vault ensuring master encryption keys remain entirely isolated inside user browser memory, mitigating cloud database leak risks.
- Cryptographic Engineering:** Implemented Web Crypto API primitives utilizing AES-GCM (256-bit) payload encryption and PBKDF2 key derivation over 100,000 hashing iterations for key security.
- Performance & Security:** Developed high-throughput Express.js REST API sync endpoints with JWT authentication and strict CORS headers, keeping vault database sync latency under 200ms.

TECHNICAL SIMULATIONS & VIRTUAL EXPERIENCES

Commonwealth Bank (CommBank) - Software Engineering Virtual Simulation

June 2026

- Problem & Scope:** Diagnosed and resolved silent data overwrite bugs across high-traffic C#/.NET Core financial API controllers handling MongoDB document updates across financial services endpoints.
- Action Taken:** Extended C# Web API controllers with `$set` atomic operators for partial payload updates and modernized an interactive React/Redux Goal Manager UI component for banking clients.
- Engineering Outcome:** Authored comprehensive automated test suites using xUnit and Moq, covering complex boundary conditions to guarantee high reliability and transactional consistency.

Y Combinator (YC Startup - Shiptivity) - Software Engineering Virtual Simulation

June 2026

- Problem & Scope:** Fixed key constraint collisions and slow re-rendering performance on a real-time drag-and-drop Kanban workflow board used by engineering teams to track task state transitions.
- Action Taken:** Architected dynamic SQLite priority reordering logic using atomic transactions to update priority ranks sequentially and integrated Dragula drag-and-drop state hooks in React.
- Engineering Outcome:** Reduced UI component re-render cycles by 30% and patched OpenSSL compilation bottlenecks in legacy Node.js Webpack configurations to optimize runtime builds.

Walmart USA - Advanced Software Engineering Virtual Simulation

May 2026

- Problem & Scope:** Solved indexing overhead and performance bottlenecks in high-volume retail inventory processing queues handling unnormalized data ingestion from legacy spreadsheets.
- Action Taken:** Implemented a generic K-ary Max Heap structure in Java using fast bitwise shift operators (`<<`, `>>>`) and built 3NF database schemas with design pattern abstractions.
- Engineering Outcome:** Accelerated queue indexing calculations by 35% and engineered automated Python ETL pipelines using `csv` and `sqlite3` to clean and consolidate disparate spreadsheets.

EDUCATION

Bachelor of Computer Applications (BCA) - Amity University Noida

Expected July 2028

- Relevant Coursework:** Data Structures & Algorithms (DSA) • Database Management Systems (DBMS) • Operating Systems (OS) • Computer Networks • Object-Oriented Programming (OOP)